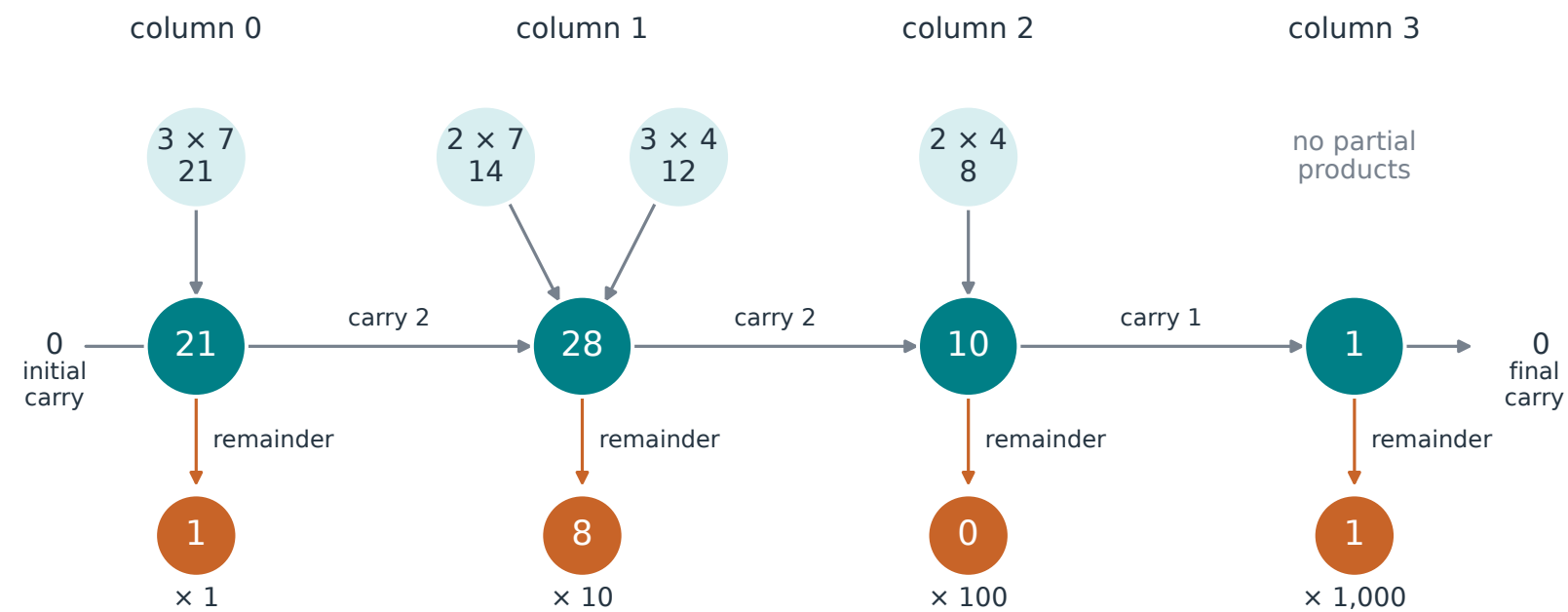


Follow the partial products and carries: $23 \times 47 = 1081$ (base 10)



Recovered product: $1 \times 1 + 8 \times 10 + 0 \times 100 + 1 \times 1000 = 1081$

For the 4096-bit construction: $B = 2^{64}$, carries $< 2^{70}$, column sides $< 2^{135} < P$.